

MATH141 - Foundations of Engineering Mathematics

Supplementary Exercises

Strand 1 - Fundamentals

1. Express the following as simply as possible.

- (a) $2c^0$ (b) $a^x \times g^x$
 (c) $b^x - b^y$ (d) $(2^3)^3$
 (e) $(ac^2)^3 \div (a^4c)^2$ (f) $x^{2a+b} \times x^{a-b}$

2. Without using a calculator, write down the values of

- (a) 9^2 (b) $9^{1/2}$
 (c) 9^{-2} (d) $(25^{-3/4})^2$
 (e) $\left(\frac{4}{9}\right)^{1\frac{1}{2}}$ (f) $(0.008)^{1\frac{1}{3}}$
 (g) $(4x^{-8})^{3/2}$ (h) $3^0 \times 2^{-2}$
 (i) $\sqrt[3]{\frac{x^9}{27}}$

3. Given that $x = a^2b$ and that $y = a\sqrt[3]{b}$, express b in terms of x and y .

4. Simplify without using a calculator:

- (a) $(-6\sqrt{5})^2$ (b) $3\sqrt{6} \times 2\sqrt{10}$
 (c) $\sqrt{15} \div \sqrt{3}$ (d) $2\sqrt{3} \div \sqrt{4}$
 (e) $\sqrt{2} + \sqrt{8}$ (f) $2\sqrt{175} - 5\sqrt{63}$

5. Find the values of the pronumerals.

- (a) $\sqrt{x} = \sqrt{72} - \sqrt{8}$
 (b) $3\sqrt{12} \times 3\sqrt{15} = y\sqrt{5}$
 (c) $\sqrt{125} - \sqrt{x} = 2\sqrt{5}$
 (d) $a + b\sqrt{2} = (2\sqrt{2} - 1)(3\sqrt{2} + 1)$
 (e) $(6 - \sqrt{3})(6 + \sqrt{3}) = c$
 (f) $(\sqrt{3} + 1)^2 = a + b\sqrt{3}$
 (g) $(2\sqrt{7} - 3\sqrt{2})^2 = x + y\sqrt{14}$

6. Simplify the following.

- (i) $\frac{1}{\sqrt{7}}$
 (ii) $\sqrt{8} \times \sqrt{6}$
 (iii) $(\sqrt{6} + \sqrt{2})^2$
 (iv) $\sqrt{288}$
 (v) $3\sqrt{45} - \sqrt{20} + 7\sqrt{5}$
 (vi) $3\sqrt{32} + 2\sqrt{75} - 5\sqrt{162}$

7. Simplify the following.

- (i) $\frac{1}{\sqrt{7} + \sqrt{3}}$ (ii) $\frac{\sqrt{7} - \sqrt{3}}{\sqrt{7} + \sqrt{3}}$
 (iii) $\frac{3}{\sqrt{5} - 2}$ (iv) $\frac{1}{5 + 3\sqrt{2}}$

8. Express with positive indices.

- (i) $a^{1/2} \times a^{-1}$
 (ii) $1 \div a^{-1/2}$
 (iii) $a^{-2}x^{-1} \div xa^{-1}$
 (iv) $\sqrt[3]{a} \times \sqrt[4]{a} \div \sqrt{a}$

(v) $\left(\frac{a^3}{b^2}\right)^5$

9. Find numerical values of

- (i) $32^{2/5}$ (ii) $4^{-5/2}$
 (ii) $125^{2/3}$

10. Write down the numbers whose logarithms to base 49, 3, 5, 8, $\frac{1}{2}$ are $\frac{1}{2}$, -3 , 1 , $\frac{1}{3}$, 2 respectively.

11. If $\log_b N = x$, write, in terms of x ,

- (a) $\log_b \left(\frac{1}{N}\right)$ (b) $\log_b(1)$
 (c) $\log_b(\sqrt{N})$ (d) $\log_b(N^{11})$

12. If $\log_2 3 = p$, $\log_2 5 = q$, write, in terms of p , q

- (a) $\log_2 25$ (b) $\log_2 15$
 (c) $\log_2 1.8$ (d) $\log_2 1$
 (e) $\log_2 150$ (e) $\log_2 \left(\frac{1}{30}\right)$

13. (i) Simplify

$$\log 4.5 + \log 70 + \log 36$$

$$-2\log 3 - \log 126.$$

[In this question $\log x + \log_{10} x$.]

(ii) Evaluate

$$\log_4 \sqrt[5]{64 \sqrt[4]{128 \sqrt{32 \times 16^{-3/4}}}}$$

14. Solve the following equations for y in terms of x .

- (i) $\ln y = -2x$
 (ii) $\ln y = \ln c + \ln x$
 (iii) $x + \ln y = \ln(x + y)$
 (iv) $\ln y = -\frac{x}{RC} + \ln \frac{E}{R}$

15. Factorise the following.

- (i) $a^2 - ax$
 (ii) $3x^3 - x^2 + x$
 (iii) $17a^2x^5 + 2a^4x^3$
 (iv) $a^2 + ab + ac + bc$
 (v) $5a + ab + 5b + b^2$
 (vi) $2ax + ay + 2bx + by$
 (vii) $3ax - bx - 3ay + by$
 (viii) $x^4 + x^3 + 2x + 2$
 (ix) $a^2x + abx + ac + aby + b^2y + bc$
 (x) $x^2 + 3x + 2$
 (xi) $x^2 + 6x + 9$
 (xii) $x^2 - 7x + 12$
 (xiii) $x^2 - 18x + 56$
 (xiv) $x^2 - 20xy + 75y^2$
 (xv) $88 + 19x + x^2$
 (xvi) $x^2 - x - 2$
 (xvii) $x^2 - x - 6$

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- (xviii) $x^2 + x - 6$
 (xix) $x^2 + 4x - 21$
 (xx) $x^2 + ax - 42a^2$
 (xxi) $110 - x - x^2$
 (xxii) $2x^2 + x - 28$
 (xxiii) $6x^2 - 7x - 3$
 (xxiv) $3x^2 + 7x + 4$
 (xxv) $x^2 - 4$
 (xxvi) $9 - 4a^2$
 (xxvii) $p^2q^2 - 100a^4$
 (xxviii) $(a+b)^2 - c^2$
 (xxix) $1 - (a-b)^2$
 (xxx) $(x+y)^2 - x^2$
 (xxxi) $x^2 + 2xy + y^2 - a^2$
 (xxxii) $a^2 - 2ab + b^2 - c^2 - 2cd - d^2$
 (xxxiii) $x^3 - y^3$
 (xxxiv) $64 + y^3$
 (xxxv) $x^6 - y^6$
 (xxxvi) $a^3 + b^3 + a + b$
 (xxxvii) $x^4y - x^2y^3 - x^3y^2 + xy^4$
 (xxxviii) $a^{12} - b^{12}$

16. Simplify

- (a) $\frac{3}{x} + \frac{4}{y}$ (b) $\frac{5}{x} \times \frac{15}{y}$
 (c) $\frac{5}{x} \div \frac{15}{y}$ (d) $\frac{1}{x^2} - \frac{1}{y^2} + 1$
 (e) $\frac{x - \frac{1}{x}}{1 + \frac{1}{x}}$ (f) $\frac{1 - \frac{y^2}{x^2}}{1 + \frac{y^2}{x^2}}$
 (g) $\frac{\frac{1}{x} - \frac{2}{x^2} - \frac{3}{x^3}}{\frac{9}{x} - x}$ (h) $\frac{2x^2 - x - 6}{\frac{4}{x^2} - 1}$
 (i) $\left(\frac{a^3 - b^3}{a - b} - \frac{a^3 + b^3}{a + b} \right) \div \frac{4ab}{a^2 - b^2}$

17. Simplify

- (a) $\frac{x-1}{2} + \frac{x+3}{5} + \frac{x+7}{10}$
 (b) $\frac{2a^2 - b^2}{a^2} - \frac{b^2 - c^2}{b^2} - \frac{c^2 - a^2}{c^2}$
 (c) $\frac{1}{x+2} + \frac{1}{x+3}$
 (d) $\frac{1}{x(x-y)} + \frac{1}{y(x+y)}$
 (e) $\frac{3}{x^2 - 4} + \frac{1}{(x-2)^2}$
 (f) $\frac{xy}{25x^2 - y^2} + \frac{2x^2y}{10x^2y + 2xy^2}$
 (g) $\frac{1}{x^2 - 9x + 20} + \frac{1}{x^2 - 11x + 30}$
 (h) $\frac{2x^2 - x - 1}{x-3} - \frac{6x^2 - x - 2}{x+3} + \frac{1}{x-1}$
 (i) $\frac{x-3}{x+2} - \frac{x-2}{x+3} + \frac{1}{x-1}$
 (j) $\frac{x-3}{x-4} - \frac{x+4}{x+3} - \frac{5}{x^2 - 16}$
 (k) $\frac{a}{(a-b)(a-c)} + \frac{b}{(b-c)(b-a)} + \frac{c}{(c-a)(c-b)}$

$$(l) \quad \frac{a}{(a-b)(a-c)} + \frac{c}{(b-c)(b-a)} + \frac{a}{(c-a)(c-b)}$$

18. Solve the following equations.

- (i) $\frac{x-2}{3x+4} = \frac{2x-3}{x+1}$
 (ii) $10x^2 + 11x = 6$

19. Solve the following equations.

- (i) $9x^{-4} + 4 = 13x^{-2}$
 (ii) $y^{2/3} + 2y^{1/3} = 8$

20. (a) Without solving, decide whether the roots of the following equations are real or not real.
 (b) In each case, write down the sum of the roots, and the product of the roots, by inspection.

- (i) $x^2 - 6x - 11 = 0$
 (ii) $2x^2 + 3x + 5 = 0$
 (iii) $4x^2 - 12x + 9 = 0$
 (iv) $5x^2 + 8 = 0$

21. Find the equation of the straight line through (1, 2), (-1, 3).

What is its gradient?

What are its intercepts on the axes?

22. (a) What is the equation of the line through (2, 3) parallel to the line $3x - 7y = 0$?

(b) What is the equation of the line through (1, -4) perpendicular to the line $4x + y + 9 = 0$?

(c) Prove that the line joining (1, 2) and (-5, 6), is perpendicular to the line joining (-2, 3) and (2, 9).

23. (a) What is the mid-point of the interval joining (4, 5) and (-2, 11)?

(b) Find the point at which the interval joining (1, 3), (9, -7) is divided in the ratio 3 : 2.

24. If A, B, C are the points (7, 1), (-3, 2), (-4, -8) respectively, show that $\triangle ABC$ is right-angled

- (a) by finding a pair of perpendicular sides, and
 (b) by verifying that Pythagoras' theorem is satisfied.

Has the triangle any other special property?

25. Show that the point (1, 2) is equidistant from the lines $3x + 4y - 8 = 0$ and $4x - 3y - 1 = 0$. Plot the point (1, 2) and sketch the lines.

26. Verify the following statements using the sum of angles formulae.

- (a) $\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$
 (b) $\cos(\pi - \theta) = -\cos \theta$
 (c) $\tan(\pi + \theta) = \tan \theta$
 (d) $\sin(0 - \theta) = -\sin \theta$

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27. Verify the following identities:

- (a) If $0 < \theta < \frac{\pi}{2}$, then $\cos \theta = \frac{1}{\sqrt{1 + \tan^2 \theta}}$
 (b) $(\sin \theta + \cos \theta)^2 = 1 + \sin 2\theta$
 (c) $\frac{1 - \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 + \sin \theta}$
 (d) $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$
 (e) $\frac{\cos 2\theta}{\cos \theta + \sin \theta} = \cos \theta - \sin \theta$.

28. (a) Use the sum of angle formula to show that

$$\begin{aligned} \cos \left(\frac{5\pi}{12} \right) &= \cos \left(\frac{\pi}{4} + \frac{\pi}{6} \right) \\ &= \frac{\sqrt{2}}{4} (\sqrt{3} - 1). \end{aligned}$$

- (b) Find the exact value of $\sin \frac{\pi}{8}$.
 (c) If $\sin \theta = \frac{5}{13}$, $\cos \theta = -\frac{12}{13}$, and $\frac{\theta}{2}$ is in the 1st Quadrant, show that

$$\sin \frac{\theta}{2} = \frac{5\sqrt{26}}{26}, \quad \cos \frac{\theta}{2} = \frac{\sqrt{26}}{26},$$

$$\tan \frac{\theta}{2} = 5.$$

 (d) If $\sin \theta = \frac{3}{5}$ and θ is in the 2nd Quadrant find $\sin 2\theta$, $\cos 2\theta$ and $\tan 2\theta$.
 (e) Write the following products as sums.
 $\cos \theta \cos 3\theta$, $\sin 5\theta$, $\cos 3\theta$.

29. Use the sum of angles formulae and the double angle formulae to show that

$$\begin{aligned} \sin 3\theta &= 3 \sin \theta - 4 \sin^3 \theta, \\ \cos 3\theta &= 4 \cos^3 \theta - 3 \cos \theta, \text{ and} \\ \tan 3\theta &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}. \end{aligned}$$

30. (i) Show that

$$\sin(x + y) \sin(x - y) = \sin^2 x - \sin^2 y.$$

(ii) By writing

$$8x = 5x + 3x \text{ and}$$

$$2x = 5x - 3x,$$

or otherwise, show that

$$\sin 8x \sin 2x = \cos^2 3x - \cos^2 5x.$$

31. Find x on the interval $0 \leq x < 2\pi$, such that

- (i) $\cos x = -\frac{1}{2}$ (ii) $\cos 2x = -\frac{1}{2}$
 (iii) $\cos x = \sin 2x$ (iv) $\sin x = \sin 2x$
 (v) $\sin x = \sin 3x$ (vi) $\cos x = \cos 2x$

32. Use the law of cosines to solve for the remaining side of $\triangle ABC$, if

- (a) $b = 12$, $c = 27$, $A = \frac{\pi}{3}$, and
 (b) $a = 4$, $b = 11$, and $C = \frac{\pi}{4}$.

33. A satellite travelling in a circular orbit 1600 km above Earth is due to pass directly over a tracking station at noon. Assume that the satellite takes 2 hours to make an orbit and that the radius of Earth is 6400 km. Find the distance between the satellite and the tracking station at 12.20 pm.

34. Use the law of sines to solve for the remaining sides of $\triangle ABC$, if $c = 10$, $A = \frac{\pi}{4}$, and $B = \frac{\pi}{3}$.

35. A satellite travelling in a circular orbit 1600 km above Earth is due to pass directly over a tracking station at noon. Assume that the satellite takes 2 hours to make an orbit and that the radius of Earth is 6400 km. If the tracking antenna is aimed $\frac{\pi}{3}$ radians above the horizon, at what time will the satellite pass through the beam of the antenna?

36. Find the area of the triangle ABC if $a = 5$, $b = 6$, and $C = \frac{\pi}{6}$.